CLAIMPROMOTEBENCH: SYNTHETIC POLICY-CONSTRAINED EVALUATION OF EVIDENCE-STATE TRANSITIONS

Anonymous authors
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ABSTRACT

Evidence-producing systems must also decide whether recorded evidence permits a claim-state change. We introduce CLAIMPROMOTEBENCH, a synthetic conformance test for one author-defined evidence-transition policy. Each controlled pair changes one policy-relevant field; scoring retains service, syntax, schema, and semantic failures in the scheduled denominator. In a frozen 300-item hidden realization, one Qwen service configuration returned 272 valid decisions and achieved 258/300 item and 113/150 pair correctness end to end (conditional: 258/272 and 113/123). A separately frozen 180-item public challenge balances its 90 non-`ALLOW` members across `HOLD`, `DEMOTE`, and `ESCALATE` (30 each). Qwen and Flash configurations achieved end-to-end item correctness of 132/180 and 90/180; pair correctness was 48/90 and 13/90. `DEMOTE` recall was 4/30 and 0/30. The deterministic executor generates the labels and defines the policy ceiling; the learned Naive Bayes comparator is weak. The artifact supports reproducible diagnosis of policy conformance, not naturalistic scientific validity or a causal model comparison.

1 INTRODUCTION

Evidence artifacts support different state changes. A finite computation covers its registered domain; a proof certificate covers an encoded statement in a pinned environment; an independent reproduction changes provenance. Treating these records as interchangeable can elevate a claim beyond the scope of its evidence.

We isolate the decision between evidence production and claim-state assignment:

> *Given a claim, typed evidence, a current state, and a requested transition, which action conforms to a fixed, machine-checkable policy?*

The benchmark treats this question as synthetic policy conformance under one authored operational definition. Label agreement measures execution of that definition. The task does not assess scientific truth, community consensus, or general research-agent competence.

CLAIMPROMOTEBENCH is a diagnostic for model-mediated routes that already sit inside larger agent systems. It asks whether such a route can parse a published contract and typed ledger, then emit the prescribed structured action. A production system can enforce this policy with the deterministic executor; our experiments measure when learned routes fail to reproduce that known decision. The executor is therefore the label generator and definitional ceiling, not a competing learned method.

The state separates research status, computational support, formal certification, and independent review. A controlled pair fixes the claim and requested transition while changing one registered prerequisite. We score `ALLOW`, `HOLD`, `DEMOTE`, and `ESCALATE`. Coverage records whether a route returns a valid decision; conditional accuracy scores that subset; end-to-end item and pair accuracy retain every scheduled case.

Our contributions are:

1. an executable controlled-pair protocol for four evidence-state actions under one fixed authored policy;

2. failure-aware item, pair, and action estimands applied to a frozen hidden realization and a pair-balanced, non-ALLOW-balanced public challenge; and

3. a replayable artifact with item-level records, privacy-filtered derivatives, executable checks, and formal fixtures.

2 Related Work

Behavioral tests and specification conformance. CheckList and contrast sets probe capabilities and local decision boundaries through targeted changes (Ribeiro et al., 2020; Gardner et al., 2020); metamorphic testing specifies relations that transformed inputs should preserve (Chan et al., 1998). Decision Model and Notation standardizes executable decision tables (Object Management Group, 2024), while ioco-style testing defines whether observed traces conform to a labeled transition specification (Tretmans, 1996). ClaimPromoteBench combines controlled perturbations with a complete policy executor. Most pairs should change the action; separate identifier, ordering, and irrelevant-evidence transformations should preserve it.

Structured output, tool calls, and abstention. Incremental parsing and grammar-constrained decoding restrict output form (Scholak et al., 2021; Geng et al., 2023). JSONSchemaBench measures schema coverage and constrained-decoding behavior (Geng et al., 2025); BFCL evaluates function selection and argument construction (Yan, 2025). These evaluations stop short of ClaimPromoteBench's cross-field evidence semantics. Selective prediction instead studies risk as a model abstains (Geifman & El-Yaniv, 2017; 2019). We separate route coverage from conditional correctness and count an absent or invalid decision as an end-to-end error. HOLD remains a valid policy action rather than a service abstention.

Rules, research agents, and evidence records. RuleArena evaluates natural-language rules across airline baggage fees, NBA transactions, and tax regulations (Zhou et al., 2025); OSVBench asks models to generate formal specifications (Li et al., 2026). ScienceAgentBench, CORE-Bench, and POPPER test broader research workflows (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). Evidence-Ledger Adjudication and LEDGER study externally labeled support or reconstructed provenance (Chen et al., 2026; Kim et al., 2026). ClaimPromoteBench narrows the target to one state-transition policy after evidence registration.

Table 1: Nearest evaluation families. "External" means that the oracle is not generated by the same policy executor shown to the evaluated route.

Setting	Input	Oracle or truth source	Policy scope	Failed output
Behavioral / contrast	Task examples plus controlled changes	Expected behavior or annotations	Capability-specific	Usually task-specific
Decision-table / ioco	Explicit rules or transition system	Specification relation	Multiple specifications	Conformance failure
Schema / tool call	Schema or tools plus request	Grammar, schema, or AST	Output interface	Format or argument error
This work	One policy, typed ledger, requested target	Same-policy label generator; no external truth	One authored policy	Every scheduled route or output failure

3 TASK AND EVIDENCE FLOW

Each item is $x = (c, E, s, \tau, t)$: a claim c, typed evidence E, current state s, requested target τ, and frozen evaluation time t. The state has four axes:

$$\begin{aligned} \texttt{research_status} &\in \{\texttt{PROPOSED}, \texttt{OPEN}, \texttt{RESOLVED}\}, \\ \texttt{computation_status} &\in \{\texttt{NONE}, \texttt{COMPUTATIONAL}\}, \\ \texttt{formal_status} &\in \{\texttt{NONE}, \texttt{FORMALLY_CERTIFIED}\}, \\ \texttt{review_status} &\in \{\texttt{UNREVIEWED}, \texttt{INDEPENDENTLY_VERIFIED}\}. \end{aligned}$$

The public policy checks dependencies, scope, freshness, provenance, runtime completion, contradictions, statement identity, and artifact integrity. Its deterministic label generator $\pi_P(x)$ evaluates the following rows from top to bottom; the first matching row fixes both the action and proposed state.

Action	First-match condition	Proposed state
DEMOTE	A currently elevated axis has an absent, broken, stale, mismatched, or policy-nonconforming recorded basis.	Lower each affected axis.
ESCALATE	A fresh, identity-matched, independently produced contradiction is open and passes its runtime and verifier checks.	Keep current state.
HOLD	The requested promotion lacks a complete valid dependency closure or any required scope, freshness, runtime, verifier, provenance, or axis-specific check.	Keep current state.
ALLOW	None of the preceding conditions holds.	Accept target state.

The model receives the complete versioned policy contract, a canonical JSON evidence ledger, and passive verifier checks that report raw facts but no transition decision. Action precedence is DEMOTE $>$ ESCALATE $>$ HOLD $>$ ALLOW; reason codes follow the policy's published total order. The response must be one JSON object with exactly six fields: `action`, `proposed_state`, `p_admissible`, `used_evidence_ids`, `reason_codes`, and `missing_evidence_ids`. No rationale text is requested.

Let $v(x)$ indicate a syntactically, schema-, and semantically valid response and let $\hat{y}(x)$ be its action. End-to-end item correctness is

$$C_{\text{item}}(x) = \mathbb{1}[v(x) = 1]\mathbb{1}[\hat{y}(x) = \pi_P(x)].$$

For controlled pair $i = (x_i^a, x_i^b)$, $C_{\text{pair}}(i) = C_{\text{item}}(x_i^a)C_{\text{item}}(x_i^b)$. Decision coverage is the fraction with $v = 1$; pair coverage is the fraction with both members valid. Conditional accuracies divide correct items or pairs by those coverage denominators. End-to-end accuracies divide by all scheduled items or pairs.

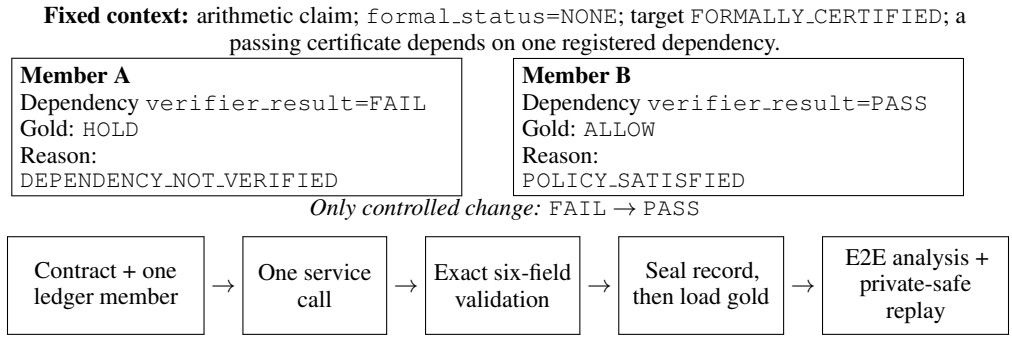


Figure 1: A concrete controlled pair and the evidence flow. The public challenge pair changes only `/evidence/1/verifier_result`; fixed identifiers and hashes are omitted from the diagram. End-to-end analysis retains every terminal failure.

Within a controlled pair, only registered fields relevant to one factor change. The illustrated member changes a dependency check from FAIL to PASS; other mappings send a missing dependency

to HOLD, a current-state binding failure to DEMOTE, and an active contradiction to ESCALATE. Separate invariance tests rename identifiers, reorder evidence, or add irrelevant records and require both the oracle and rendered prompt semantics to remain unchanged.

Table 2: Factor inventory, grouped by policy family. Labels humanize the machine-readable factor identifiers; counts are 20 for r6 and 15 for the challenge.

Family	r6 factors	Public-challenge factors
Dependency	dependency statement, dependency verifier, direct dependency, transitive dependency	dependency verifier
Formal identity	claim identity, statement identity, trust base	—
Numerical, current basis	domain, precision, preregistration, repetitions	current computation binding, staleness, verifier; current review environment, identity; repetitions
Provenance	cold start, review environment, reviewer identity	reviewer identity
Runtime	run completion, tool exit, verifier execution	tool exit
Freshness, contradiction	age, contradiction, current binding	age; contradiction claim identity, freshness, reviewer identity, status, verifier execution

4 Benchmark and Evaluation Design

4.1 Synthetic benchmark

Version 1 contains 200 pairs (400 items): 50 public-development pairs and 150 hidden pairs over the 20 factors in Table 2. Numerical cases recompute registered finite-grid values and scope, while formal cases compile generated Lean sources under a pinned toolchain. These checks establish internal policy consistency, not external scientific validity.

The fresh r6 realization contains 150 hidden pairs and 300 unique prompts from the same ontology and mostly the same template family. A construction commitment and request schedule were frozen before inference; all 600 terminal cells for two service lanes were sealed before gold loading. Each item received one call with zero client retries. The gold distribution is imbalanced: 150 ALLOW, 134 HOLD, eight DEMOTE, and eight ESCALATE items. We also froze a four-class multinomial Naive Bayes classifier ($\alpha = 1$) trained on identifier-normalized unigram and bigram prompt counts from the 100 public-development items. This text classifier is a low-cost reference rather than a competitive learned baseline.

4.2 Pair-balanced public robustness challenge

The separately precommitted public challenge tests action asymmetry over the 15 factors in Table 2, with six pairs per factor. Each of its 90 pairs contains one ALLOW member. The other members are balanced across HOLD, DEMOTE, and ESCALATE (30 each), yielding gold support of 90/30/30/30 rather than four-class balance. We froze the dataset, labels, prompts, adapters, analysis, and execution contract before inference. Each service route made 180 calls with zero client retries. This known public construction is a robustness test for the fixed policy, not an estimate of hidden-test or natural-distribution generalization.

The two routes differ in model and service stack, including output-constraining behavior. We report descriptive service-configuration outcomes because the design cannot identify the effect of model weights, provider, schema mode, or any single component.

5 Results

5.1 Coverage, conditional accuracy, and end-to-end accuracy

The r6 Qwen configuration returned 272/300 valid decisions and 123/150 complete pairs (Table 3). It answered 258/300 items and 113/150 pairs correctly end to end; conditional correctness was 258/272 and 113/123. The 28 invalid items comprise 15 HTTP errors, ten timeouts, and three semantic failures. The Naive Bayes reference answered 184/300 items and 34/150 pairs correctly and did not recover either rare action reliably.

Table 3: Coverage and correctness with explicit denominators. E2E columns retain all scheduled cases; conditional columns use valid decisions or complete pairs. N/E denotes an empty conditional denominator.

Configuration	Valid items	E2E item	Cond. item	Complete pairs	E2E pair	Cond. pair
r6 Qwen3.7-Max	272/300	258/300	258/272	123/150	113/150	113/123
r6 public-dev text NB	300/300	184/300	184/300	150/150	34/150	34/150
Old Flash service-interface failure	0/300	0/300	N/E	0/150	0/150	N/E
Public challenge Qwen	148/180	132/180	132/148	64/90	48/90	48/64
Public challenge Flash	146/180	90/180	90/146	61/90	13/90	13/61

The old r6 Flash service interface returned HTTP 401 for all 300 requests and produced no model output. Its route-level end-to-end counts are 0/300 items and 0/150 pairs; model efficacy is non-evaluable. The later challenge route produced outputs through a different service stack. That result establishes executability of the later route and cannot recover the missing r6 estimate or support a causal comparison.

On the pair-balanced challenge, Qwen returned 148/180 valid decisions and 64/90 complete pairs; Flash returned 146/180 and 61/90. Their end-to-end item/pair accuracies were .7333/.5333 and .5000/.1444, compared with conditional accuracies of .8919/.7500 and .6164/.2131. Qwen recorded 31 transport failures and one semantic failure; Flash recorded 34 semantic failures. Pair correctness falls when either member is invalid or wrong, which explains the larger gap between conditional and end-to-end pair estimates.

5.2 Factor-level pair outcomes

Factor-level pair results vary sharply (Table 4). R6 Qwen has two zero-accuracy factors and seven perfect factors; dependency statement yields 0/7 correct pairs from 2/7 complete, and current binding yields 0/8 from 6/8. Challenge Qwen has three zero-accuracy factors and two perfect factors; challenge Flash has 11 and zero. Supports of six to eight pairs make these descriptive diagnostics. The artifact retains every factor-level numerator and denominator.

Table 4: Factor-level pair diagnostics. Coverage and E2E ranges are minima–maxima across factors; zero/perfect counts use E2E pair accuracy. The reviewer artifact contains every factor row.

Configuration	Factors	Pair coverage range	E2E pair range	Zero / perfect
r6 Qwen	20	.2857–1.0000	.0000–1.0000	2 / 7
Challenge Qwen	15	.5000–1.0000	.0000–1.0000	3 / 2
Challenge Flash	15	.0000–1.0000	.0000–.8333	11 / 0

5.3 Action recall and decision confusion matrices

Table 5 shows end-to-end recall for every configuration that produced decisions, so no-valid-decision cells count as misses. R6 has only eight `DEMOTE` and eight `ESCALATE` items. The weak Naive Bayes reference recovers neither rare action. The public challenge supplies 30 members for each non-`ALLOW` action and preserves the `DEMOTE` weakness: Qwen recovers 4/30 and Flash 0/30.

Table 5: End-to-end recall by gold action; no-valid cells are misses. Gold supports are 150/134/8/8 for r6 and 90/30/30/30 for each challenge lane. Macro-F1 averages all four actions with zero division set to zero.

Configuration	`ALLOW`	`HOLD`	`DEMOTE`	`ESCALATE`	Bal. recall	Macro-F1
r6 Qwen3.7-Max	.9533	.8209	.0000	.6250	.5998	.6265
r6 public-dev text NB	.5000	.8134	.0000	.0000	.3284	.3124
Public challenge Qwen	.8778	.8667	.1333	.7667	.6611	.7004
Public challenge Flash	.7556	.3000	.0000	.4333	.3722	.3928

Table 6 gives every end-to-end confusion cell for the four configurations that produced decisions. The old Flash interface generated no decisions and therefore has no decision confusion matrix; its route-level failure remains in Table 3. The final column distinguishes absence of a valid decision from any policy action. R6 Qwen maps all eight gold DEMOTE items to ALLOW. On the public challenge, Qwen's 20 valid gold-DEMOTE responses split across all four actions, while Flash never predicts DEMOTE.

Table 6: Decision confusion matrices for output-producing configurations. Rows are gold actions; action columns are valid predictions; "No valid" includes service, syntax, schema, and semantic failures.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	No valid
r6 Qwen	ALLOW	143	0	0	0	7
	HOLD	4	110	0	2	18
	DEMOTE	8	0	0	0	0
	ESCALATE	0	0	0	5	3
r6 text NB	ALLOW	75	75	0	0	0
	HOLD	25	109	0	0	0
	DEMOTE	8	0	0	0	0
	ESCALATE	8	0	0	0	0
Challenge Qwen	ALLOW	79	0	0	0	11
	HOLD	0	26	0	0	4
	DEMOTE	8	4	4	4	10
	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	0	8	10
	HOLD	13	9	0	0	8
	DEMOTE	25	3	0	0	2
	ESCALATE	0	3	0	13	14

5.4 Definitional policy ceiling

The deterministic policy executor reads the model-visible item and the policy used to generate its label. It reproduces every r6 and public-challenge label and pair by construction. We call this agreement a *definitional policy ceiling*. The executor is a label generator, and post-generation gates that reuse it are implementation checks. Neither supplies learned-model evidence or external validation of the authored policy.

6 Discussion

R6 shows that conditional and end-to-end estimates answer different questions. The Qwen route is accurate after a valid response exists, yet 28 invalid items and 27 incomplete pairs remain deployment failures under the frozen contract. Its class distribution also hides a complete failure on DEMOTE. The pair-balanced challenge raises each non-ALLOW support to 30 and confirms that DEMOTE remains difficult. Action recall and pair correctness expose this behavior more directly than aggregate item accuracy.

The Flash routes illustrate why service failures and model behavior must remain distinct. Three hundred HTTP 401 responses document failure of the old r6 service interface, while outputs from the later stack establish only that the later path ran. Different models, services, and adapters preclude a causal ablation.

The deterministic executor provides the reference action that makes conformance measurable and can enforce the policy in production. Learned routes remain relevant when a larger agent already uses a model to interpret typed records and emit tool-facing decisions; ClaimPromoteBench tests that boundary and shows where deterministic enforcement is warranted. The Naive Bayes result supplies a reproducible floor, not evidence against competitive learned methods.

7 Limitations

The evidence establishes internal conformance for one authored policy. R6 retains the same ontology and mostly the same template family as development data. The public challenge is known in advance

and balances only the non-ALLOW member of each pair. No external policy consensus, downstream-outcome validation, or cross-policy evaluation supports the label space. Source-level review of 12 pointer-only CORE-Bench candidates made zero model calls and created zero cases because the records lacked the time-stamped artifacts and provenance required for an author-policy mapping. No human annotators or agreement statistic are claimed. Naturalistic validity therefore remains untested.

Coverage limits every conditional estimate. R6 loses 28/300 decisions and 27/150 complete pairs; the public challenge loses 32/180 Qwen decisions and 34/180 Flash decisions. Conditional correctness therefore describes only the covered subset and must be read beside end-to-end correctness on every scheduled item or pair.

Each evaluated route executes one call per item with zero client retries. These exact frozen-corpus counts do not estimate sampling variance, retry-policy effects, temporal stability, or fixed model weights. Six to eight pairs per factor support diagnostics rather than stable factor rankings.

The learned comparison is limited to a weak Naive Bayes text classifier trained on 100 public-development items. The deterministic executor generates the labels and cannot substitute for an independently trained baseline. Stronger learned comparators, independently authored policies, repeated route configurations, and licensed natural cases would test dimensions that the current evidence leaves open.

8 Reproducibility and Artifact Policy

The repository binds policies, generators, datasets, prompts, terminal records, gold-loading order, analyses, and receipts. R6 freezes its construction and 600-request schedule before inference and seals each terminal record before gold loading. The public challenge binds its inputs, adapters, schedule, and analysis before its calls. Paper construction replays committed records and does not rerun completed singleton studies.

A post-r6 audit found provider hidden-thinking blocks in 275 source envelopes; scoring had used only final response text and parsed decisions. The reviewer derivative excludes those envelopes through a text-only allowlist and verifies source-to-derived hashes and scientific-payload equivalence. The anonymous bundle contains replayable item records, policy and prompt bindings, executable checks, formal checks, and privacy receipts. Its manifest derives the current file count from the generated tree. Fail-closed scans reject author identity, repository or workflow identifiers, provider response identifiers, hidden reasoning, and unknown record fields.

9 AI Assistance Disclosure

Generative AI tools assisted literature search, planning, code and benchmark drafting, debugging, analysis checks, and manuscript editing. The author chose the research question, defined the policy and protocol, made the study and analysis decisions, verified the benchmark, results, and references, and is responsible for the manuscript. Quantitative results come from validated study artifacts. AI systems are neither authors nor human annotators.

10 Conclusion

ClaimPromoteBench provides a replayable conformance test for one synthetic evidence-transition policy. End-to-end denominators reveal failures hidden by conditional accuracy, while pair and action metrics expose the persistent DEMOTE weakness: Qwen and Flash recover 4/30 and 0/30 cases on the public challenge. These findings support a diagnostic protocol for model-mediated policy decisions. Natural scientific judgment, causal service effects, and cross-policy generalization require new studies.

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